

Inclusive Character Creator

A Showcase of Inclusive Design Principles for 3D Character Creators

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ABSTRACT

Inclusive Character Creator is a speculative design research project that seeks to address some of the long-standing issues of sexism, racism, ableism, and sizeism prevalent in most 3D character creators in interactive media. This project focuses on stylized and expressive features rather than hyperrealism. It seeks to redefine what it means to start a character from a “default body,” a definition that usually results in creating a biased system that relies on media norms. A version 1.0 has been built, encapsulating fundamental principles resulting from research.

CCS CONCEPTS

• **Applied computing** → Arts and humanities; Media arts; • **Human-centered computing** → Human computer interaction (HCI); • **Computing methodologies** → Computer graphics; Rendering; Non-photorealistic rendering.

KEYWORDS

Character, representation, avatar, diversity, inclusion

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1 INTRODUCTION

In a 2021 study of game streaming content from the top 20 Twitch streamers, female characters, LGBTQIA+ characters, characters with a physical disability, and characters with a large body type are represented at rates far below the demographic of the U.S. population, and female characters that are represented tend to be sexualized [Geena Davis Institute on Gender in Media, 2021]. Over recent

years, however, the appearance of playable characters has begun to diversify as character creators in video games, and avatar makers in social media platforms have become more common. But even with the addition of more options, the results are often still limited. Many character creators start with the premise of a “default body,” which is usually a “non-disabled, cisgender white male who is neither young nor old and has a stereotypical body type” [Olofsson and Adolfsson, 2017]. Additional features to modify this template are then treated as secondary. Thus, character creators still exhibit representation issues in terms of gender, race, body size, disability, and more. This research explores alternative design principles that could build more inclusive character creators.

2 APPROACH

This project combined expert feedback, playtest surveys, workshops, and documentation research to develop design requirements for improving future character creators. Consultant feedback came from five individuals with expertise and/or lived experience from specific marginalized and underrepresented identities. Playtest feedback was gathered from 12 participant surveys early in the project’s prototyping phase. The workshop was a two-hour experience where 16 participants co-designed and selected hair options they would like to see in character creators. The documentation research drew from developer blogs, artist tutorials, and classification systems from the fashion and medical industry to synthesize a production process for the interactive prototype built with Maya and Unity. Feedback from the research was integrated into a demonstration user interface (UI) that showcases design principles for inclusive character creators.

3 RESULTS

The current demonstration user interface (UI) includes three main sizes: small, medium, and large, representing character body *size* and *shape*. For body *shape*, the UI features five main shapes: rectangle, pear, heart, hourglass, and oval, which define the body in relation to the shoulders, mid-section, and waist. In addition, there are underwear options for a bra, no top, briefs, shorts, and a slider to control chest size. For disability representation, this project features transhumeral and transfemoral amputation and prosthesis and a wheelchair option, and a seated pose compatible with all body types.

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Figure 1: Coily stylized hair (left), wheelchair and prosthesis (middle), and skin tone options (right) available in the prototype

The *skin tone* UI includes 45 natural skin tones ranging from deep to medium to light and then warm to neutral to cool undertones. All skin tones show palm coloring. In addition, there is an alternate skin color scheme to show that even fantasy palettes need to provide options for darker skin tones.

For *hair*, the UI includes a mid-length, loose, coily hairstyle. The hair was made with hair cards, hair helmets, and a curl mesh following a process illustrated by Campo Santo [Campo Santo, 2018] that resulted in a polycount of 2.5k.

The UI and UX center on Black and Brown identities within the feature ordering hierarchy. Hover text exposes the categorizations and descriptions used for each feature to help educate the player and hold the developer’s rationale accountable.

4 CONCLUSION AND FUTURE WORK

The *Inclusive Character Creator* 1.0 is a critical starting point that every game developer and tool creator can use as they consider inclusivity in their designs. By centering on Black, Brown, disabled, and non-binary identities, this research demonstrates a tool

through which players are less likely to experience exclusion. Tool design in computer graphics can be improved and affirm diverse identities to avoid perpetuating societal injustices. The 2.0 UI/UX will feature additional, larger body sizes, more underwear options (such as binders and undershirts), parameterization for limbs and proportions; different portrayals of physical disability and assistive tech; skin textures for conditions such as vitiligo, acne, and rosacea, as well as body hair and body markings; at least one hairstyle for each hair type according to the Andre Walker Hair Typing system [Walker, 2022], along with modular hair additions for edges, bangs, updos, and facial hair. It will be expanded to portray more aspects of identity and be used as an educational framework that game developers can use to build their contexts.

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